CSE 4345: Big Data Analytics

Week 1, Part 2 — Seminal Research & Case Study

Department of Computer Science & Engineering
Premier University, Chittagong

Semester 2026

Part 2 Overview

Today's Agenda

Seminal Papers Covered

1. Laney, D. (2001). *3D Data Management*. META Group.
2. Brewer, E. (2000). *Towards Robust Distributed Systems*. PODC.
3. Gilbert & Lynch (2002). *Brewer's Conjecture*. ACM SIGACT.

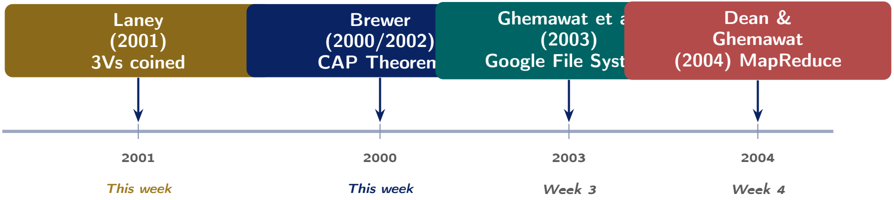
Case Study

Manyika et al. (2011). *Big Data: The Next Frontier for Innovation, Competition, and Productivity*. McKinsey Global Institute.

CLO Addressed

Research Paper 1 — Laney (2001): The 3Vs Origin

Research Landscape: Why These Papers?



Reading Sequence

Laney (2001) defines *what* Big Data is $\rightarrow$ Brewer/Gilbert (2000–2002) defines *why* distributed design is hard $\rightarrow$ GFS (2003) and MapReduce (2004) provide the *engineering solutions*. This course follows that exact intellectual lineage.

Laney (2001): Publication Context

Full Citation

Laney, D. (2001, February 6). **3D Data Management: Controlling Data Volume, Velocity and Variety.** META Group Application Delivery Strategies Research Note, File 949.

Venue: META Group analyst memo (2 pages)
Cited by: >20,000 times (Google Scholar, 2024)
Impact: Became the canonical industry and academic definition of Big Data

The Author

Doug Laney was a research analyst at META Group (later acquired by Gartner in 2005). He wrote this as a short practitioner note about **e-commerce data management challenges**—not a formal academic

Laney's Own Reflection (2012 Interview)

"It was sort of a throwaway piece. I had been thinking about the data challenges at e-commerce companies and wrote down three dimensions. I never expected it to become the definition of an entire era of computing."

—Doug Laney, Gartner Blog, 2012

Why Study a 2-page Memo?

It illustrates that **foundational frameworks** need not come from prestigious venues. The quality of the *insight* matters more than the journal. This is a recurring theme in Big Data research.

Laney (2001): The Original 3Vs—Close Reading

Opening Argument (Paraphrased)

"E-commerce has exploded the volume, velocity and variety of information that companies must manage. Retailers and their trading partners face the challenge of managing large, fast-moving and diverse data from the Web, RFID, point-of-sale sensors and order entry systems."

Laney's original definitions:

- **Volume:** Transactions at an online retailer generate **exponentially more** data than equivalent brick-and-mortar operations. Click-stream data, session logs, product views—each visit generates thousands of records.
- **Velocity:** The **time-sensitivity** of data. Stock prices, fraud signals, and real-time offers must be

The Insight That Made This Seminal

Laney was the first to observe that these three dimensions are **interdependent and compound**:

- High velocity + high variety = integration is nearly impossible in real time
- High volume + high variety = storage schemas must be flexible
- High volume + high velocity = single-node processing cannot keep up

This **3D problem space** is why no single traditional tool solves Big Data—you need an *ecosystem*.

Laney (2001): The E-Commerce Motivation

The year is 2001. The dot-com boom has just ended. What data problems were practitioners actually facing?

Laney's Observed Challenges

Dimension	Observed Problem (2001)
Volume	Amazon.com: millions of product pages, billions of click-stream events per month. No RDBMS could store it all.
Velocity	eBay: real-time auction bids must update in seconds. Batch-nightly processing was useless.

2024 Parallel: Same Problem, 1,000× Scale

- **Volume (2024):** Amazon: >1.6 million orders per day; petabytes of product images, reviews, fulfillment data
- **Velocity:** Daraz.com.bd: flash-sale order spikes of 10,000 orders/minute require real-time inventory sync
- **Variety:** Shopper GPS + purchase history + product image + chatbot transcript + payment gateway log = five different formats in one customer journey

The Enduring Relevance

Laney's 3Vs from 2001 describe a problem

Laney (2001): Legacy, Extensions, and Critique

What the paper triggered:

- Gartner adopted 3Vs as official Big Data definition (2012)
- IBM extended to **4Vs** (adding Veracity, 2013)
- IDC added **Value** as the 5th V (2014)
- Over 40 papers have proposed additional Vs (Validity, Volatility, Venue, Vocabulary. . .)

The 5Vs Consensus (Gartner / IDC, 2012–2014)

V	Added by
Volume, Velocity, Variety	Laney (2001)
Veracity (quality)	IBM (2013)
Value (utility)	IDC (2014)

Academic Critique of the 3Vs

The 3Vs are a **useful taxonomy**, not a formal theory:

- They describe *symptoms*, not solutions
- They do not specify *thresholds*: how much volume is “big”?
- They ignore *social and ethical* dimensions of data (privacy, consent, bias)
- Kitchin (2014) argues they reinforce a **techno-deterministic** view that more data always produces better insights

Reference: Kitchin, R. (2014). *The Data Revolution*. Sage. Ch. 4.

Research Paper 2 — Brewer (2000) & Gilbert & Lynch (2002): CAP Theorem

Brewer (2000): The CAP Conjecture

Full Citations

[1] Brewer, E. A. (2000). **Towards Robust Distributed Systems.** Invited Keynote, ACM Symposium on Principles of Distributed Computing (PODC), Portland, Oregon.

[2] Gilbert, S. & Lynch, N. (2002). **Brewer's Conjecture and the Feasibility of Consistent, Available, Partition-Tolerant Web Services.** ACM SIGACT News, 33(2), 51–59.

About the Authors

Eric Brewer: UC Berkeley professor; co-founder of Inktomi (early web search engine), VP of Infrastructure at Google. His *practitioner* instincts drove the conjecture.

The Conjecture (Brewer, 2000)

"It is impossible for a web service to simultaneously provide all three of the following guarantees:

- ***Consistency**: all nodes see the same data at the same time;*
- ***Availability**: a guarantee that every request receives a response;*
- ***Partition tolerance**: the system continues to operate despite arbitrary message loss or failure of part of the system."*

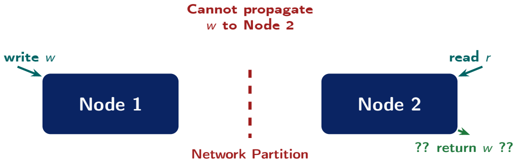
—Brewer, PODC 2000 Keynote

Gilbert & Lynch (2002): The Formal Proof

Proof Strategy: Contradiction

Gilbert & Lynch used an **impossibility argument**:

1. Assume a system guarantees C, A, and P simultaneously.
2. Construct a scenario with a network partition: Node 1 and Node 2 cannot communicate.
3. A write w is sent to Node 1. A read r is sent to Node 2.
4. **Availability** requires Node 2 to respond to r.
5. **Consistency** requires Node 2's response to reflect w.
6. But **Partition tolerance** means Node 2 cannot receive w from Node 1.
7. **Contradiction** $\Rightarrow$ C, A, P cannot all hold



Design Implication

Since partitions **cannot be prevented** in real networks, every distributed system *must* tolerate P. The only real design choice is: **C or A** when a partition occurs.

CAP Theorem: Real System Choices & Nuances

Choice	Systems	Trade-off
CP	HBase, ZooKeeper, MongoDB (default)	Returns error instead of stale data during partition
AP	Cassandra, DynamoDB, CouchDB	Returns possibly stale data; reconciles later
CA	Traditional RDBMS (MySQL, PostgreSQL)	Only works without network partitions (single node)

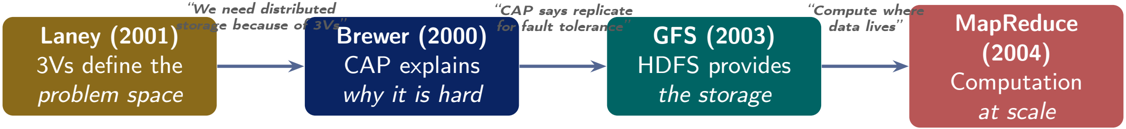
Bangladesh Example: bKash vs. Facebook

Important Nuance: Kleppmann (2015)

"Please stop calling databases CP or AP" — Martin Kleppmann, *A Critique of the CAP Theorem*, arXiv:1509.05393.

- CAP's definitions are more nuanced than binary classifications suggest
- Real systems provide **tunable consistency** (e.g., Cassandra's consistency levels: ONE, QUORUM, ALL)
- **PACELC model** (Abadi, 2012) extends CAP: even without partitions, there is a Latency vs. Consistency trade-off
- CAP is a **useful mental model**, not a precise engineering specification

Connecting the Papers: From 3Vs to Distributed Design



The Intellectual Lineage of This Course

Laney's 3Vs **motivated** distributed storage (because a single server cannot handle 3Vs simultaneously) → Brewer's CAP **constrained** the design space (you must choose C or A) → Google's GFS and MapReduce **implemented** a concrete CP-leaning architecture → Hadoop **open-sourced** those ideas. This entire arc from 2001–2006 forms the foundation of everything you will study in Weeks 2–10.

Case Study — McKinsey Global Institute (2011)

About the Report

Full Citation

Manyika, J., Chui, M., Brown, B., Bughin, J., Dobbs, R., Roxburgh, C., & Byers, A.H. (2011, May). **Big Data: The Next Frontier for Innovation, Competition, and Productivity.** McKinsey Global Institute.

Length: 156 pages
Downloaded: >2 million times
Cited by: >15,000 academic papers
Lead author: James Manyika, Director, McKinsey Global Institute

Scope

Surveyed Big Data potential across **five domains**: US healthcare, European public administration, US retail,

Why It Mattered in 2011

This was the first large-scale, cross-industry **quantified economic analysis** of Big Data's potential value. Before this report, Big Data was primarily a *technical* term used by engineers. McKinsey made it a *boardroom* term.

Historical Context (2011)

- Hadoop 1.0 had just been released (2011)
- Spark did not yet exist (2012)
- "Data Scientist" was named best job in America by HBR (2012)
- AWS was 5 years old; cloud computing was nascent

Key Quantitative Findings

Domain	Estimated
US Healthcare	$300 billion+
European Public Sector	100 billion+
US Retail (operating margin)	60%+ inc
Global Manufacturing	$600 bil
Personal location data (EU)	$100 bil
US talent shortfall (deep analytics)	**140,000–190,000 people by**
US manager shortfall (data-savvy)	**1.5 million by**

The Headline Argument

"Big data will become a key basis of competition, underpinning new waves of productivity growth, innovation, and consumer surplus. Leaders in every sector will have to grapple with the implications of big data, not just a few data-oriented managers."

—Manyika et al., 2011, p. 1

The Talent Gap Was Under-estimated

The report predicted 140,000–190,000 deep analytics jobs needed by 2018. LinkedIn's 2018 Workforce Report found **250,000+ unfilled**

Healthcare: The Flagship Case

US Healthcare: $300B+ Annual Value Potential

McKinsey identified five mechanism categories:

1. **Clinical decision support**: Mining EHR data to identify optimal treatments and reduce diagnostic errors. Potential: $165 billion.
2. **R&D optimisation**: Mining clinical trial data to reduce drug development cost and time. Potential: $70 billion.
3. **Public health**: Epidemic detection via syndromic surveillance of pharmacy sales and emergency room data. Potential: $30 billion.
4. **Fraud and waste detection**: Identifying anomalous billing patterns in Medicare/Medicaid claims. Potential: $17 billion.

2024 Reality Check: What Was Achieved?

Largely realised:

- Google DeepMind's AlphaFold (2020) — solved protein structure prediction using big data + deep learning
- CMS fraud detection using ML saves $4B/year (2022 GAO report)
- COVID-19 genomic surveillance systems (GISAID) directly applied big data to track variants in real time

Underachieved:

- EHR interoperability remains poor (data silos still exist)
- **HIPAA compliance** slows data sharing;

Critical Analysis: What Did McKinsey Get Right and Wrong?

What McKinsey Got Right ✓

- **Talent gap**: understated, not overstated
- **Competitive differentiation**: tech companies that used big data (Netflix, Amazon, Google) dominated their industries
- **Healthcare value**: clinical AI has validated many of the predicted gains
- **Retail personalisation**: Amazon's recommendation engine drives $\approx$35% of revenue—enabled by big data
- **Urgency**: The hiring and investment wave that followed this report built the infrastructure now used for AI

What McKinsey Missed ×

- **Privacy & regulation**: No mention of GDPR (2018), CCPA, or the political cost of mass data collection (Cambridge Analytica, 2018)
- **Algorithmic bias**: Big data models can *amplify* discrimination (race, gender, credit scoring)
- **Google Flu Trends cautionary tale (2008–2015)**: Hyped as proof of big data power; failed catastrophically due to **overfitting and Goodhart's Law**
- **Cloud dominance**: Assumed on-premise Hadoop clusters; actual market moved to AWS/GCP/Azure
- **Deep learning**: Report assumed statistical

Cautionary Interlude: Google Flu Trends (2008–2015)

The Claim (Ginsberg et al., 2009, *Nature*)

"We can accurately estimate influenza activity in each region of the United States, up to two weeks ahead of traditional CDC reports, simply by tracking the volume of flu-related Google search queries."

Initial results: Predicted flu activity with $r^2 = 0.97$ correlation to CDC data. Widely cited as proof that big data could replace traditional surveillance.

What Went Wrong

- **Overfitting**: 45 query terms selected post-hoc from 50 million possibilities
- **Goodhart's Law**: When a measure becomes a target, it ceases to be a good measure. Media

Lessons for Big Data Practitioners

1. **Correlation $\neq$ causation**: A high r^2 on training data does not guarantee real-world validity
2. **Feedback loops**: Human behaviour changes when people know they are being measured
3. **Ground truth matters**: Big data **supplements**, it does not replace, rigorous data collection
4. **Reproducibility**: Selecting 45 features from 50M candidates without cross-validation is a **data dredging** error

Reference: Lazer et al. (2014). *The Parable of Google Flu: Traps in Big Data Analysis.*

Bangladesh & South Asia Relevance

McKinsey's Emerging-Economy Blind Spot

The 2011 report focused exclusively on the USA and Europe. But the McKinsey logic applies directly to Bangladesh:

Sector	Big Data Opportunity
Mobile finance	bKash: 65M+ accounts, real-time fraud, credit scoring for the unbanked
Garment (RMG)	Supply-chain optimisation, energy prediction in 4,000+ factories
Agriculture	Crop yield prediction from satellite data + weather APIs

A2i & ICT Division Initiatives

Bangladesh's **Aspire to Innovate (a2i)** programme is actively building national data infrastructure:

- **National Data Warehouse**: Integrating 49 ministries' databases
- **NID biometric database**: 110 million records used for KYC in banking and telecoms
- **Health Bulletin DHIS2**: District-level health data aggregation
- **Bangladesh Open Data Portal**: data.gov.bd (launched 2021)

Each of these is a real-world manifestation of the McKinsey public sector big data thesis

Guided Discussion & Live Exercise

Class Discussion Questions

Instructions

Discuss in pairs (5 minutes), then share with class. There are no single correct answers—justify your position with evidence from today's papers.

D1. Laney wrote his 3Vs memo in 2001 to describe e-commerce challenges. Do you think the **same three dimensions** are still the most important way to characterise Big Data challenges in 2024, or has the field evolved past this framework? What would you change or add?

D2. The McKinsey report predicted massive economic value from Big Data, but also **missed** the privacy, bias, and regulatory implications entirely. Does this mean we should be *sceptical* of similar economic projections for AI today? What should analysts include that McKinsey did not?

D3. The Google Flu Trends failure is often cited as a warning against "big data hubris." Can you identify a current example in Bangladesh where a big data system might **fail for similar reasons** (feedback loop, overfitting, missing ground truth)?

Live Exercise: Apply the Frameworks

Instructions

Individual work, 10 minutes. Submit on paper or LMS.

Exercise A: 5Vs Classification (CLO1)

For each of the following data assets belonging to a Bangladeshi e-commerce platform, rate each V on a scale of **Low** / **Medium** / **High** and **justify in one sentence**:

1. Daily product sales CSV exported from MySQL at midnight
2. Real-time GPS location of 500 delivery riders
3. Customer-uploaded product review photos
4. HL7 medical certificate required for pharmaceutical products

Exercise B: System Design (CAP)

You are the CTO of a new Bangladeshi super-app (like bKash + Shohoz + Chaldal combined). For each sub-system below, state which **CAP trade-off** (CP or AP) you would choose and why:

1. **Mobile wallet balance**: showing current taka balance
2. **Product availability**: showing whether an item is in stock
3. **Ride fare estimate**: showing the price

Full References

Seminal Papers

- Laney, D. (2001, Feb 6). **3D Data Management: Controlling Data Volume, Velocity and Variety.** META Group Research Note, File 949.
- Brewer, E. A. (2000). **Towards Robust Distributed Systems.** ACM PODC Keynote, Portland, OR.
- Gilbert, S. & Lynch, N. (2002). **Brewer's Conjecture and the Feasibility of Consistent, Available, Partition-Tolerant Web Services.** *ACM SIGACT News*, 33(2), 51–59.
- Kleppmann, M. (2015). **A Critique of the CAP Theorem.** *arXiv*:1509.05393. *[Recommended reading]*
- Abadi, D. (2012). **Consistency Tradeoffs in Modern Distributed Database System Design: CAP is Only Part of the Story.** *IEEE Computer*, 45(2), 37–42. *[PACELC]*

Case Study & Critique

- Manyika, J. et al. (2011). **Big Data: The Next Frontier.** McKinsey Global Institute.
- Ginsberg, J. et al. (2009). **Detecting influenza epidemics using search engine query data.** *Nature*, 457, 1012–1014.
- Lazer, D. et al. (2014). **The Parable of Google Flu: Traps in Big Data Analysis.** *Science*, 343(6176), 1203–1205.

End of Week 1

CSE 4345: Big Data Analytics

"The world is one big data problem."

—Andrew McAfee, MIT Sloan School

Next Week: Week 2 (Part 1 & Part 2)

Challenges of Big Data: Extraction, Integration & Heterogeneous Information

Seminal Paper: Halevy, Rajaraman & Ordille (2006) · Case Study: Healthcare Interoperability (HL7/FHIR)

Reading before Week 2: [SA] Ch. 2 · **[TE]** Ch. 3